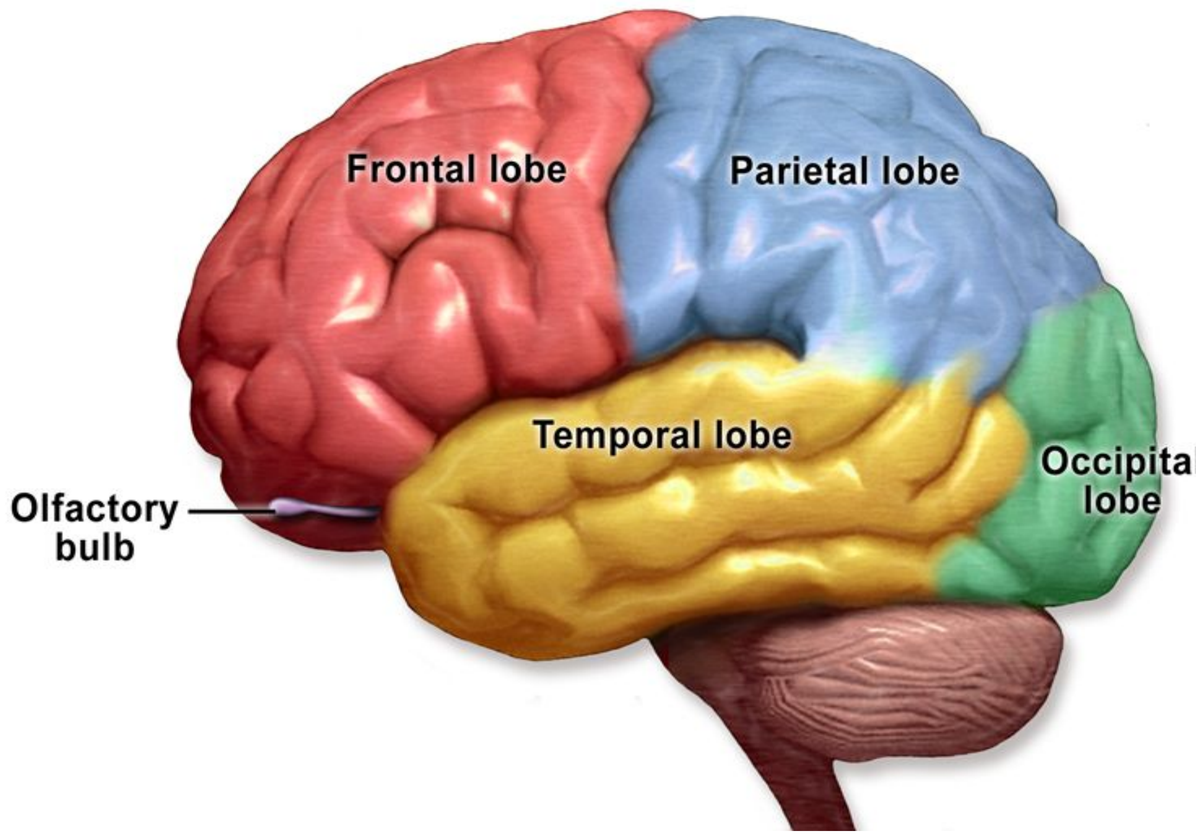
Organization and levels of Nervous system



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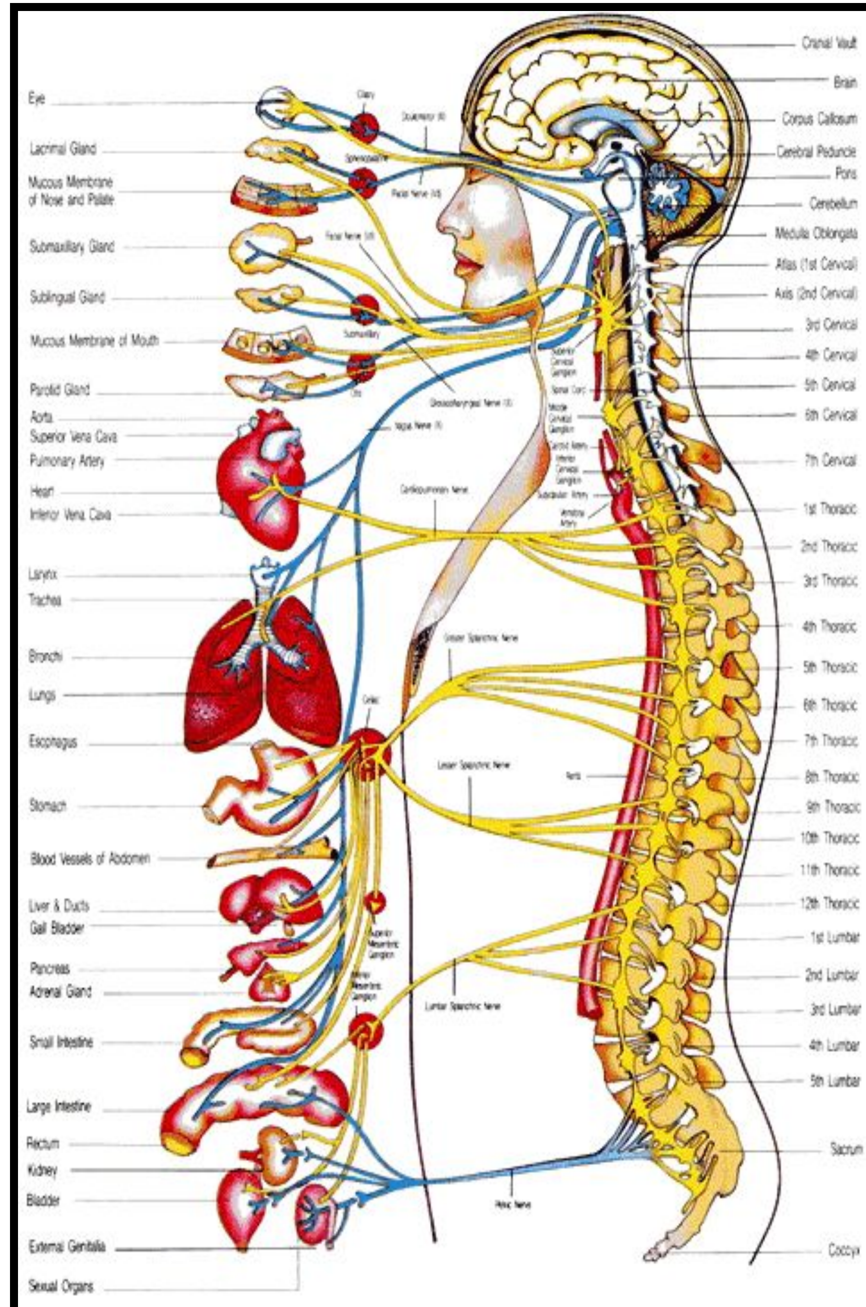
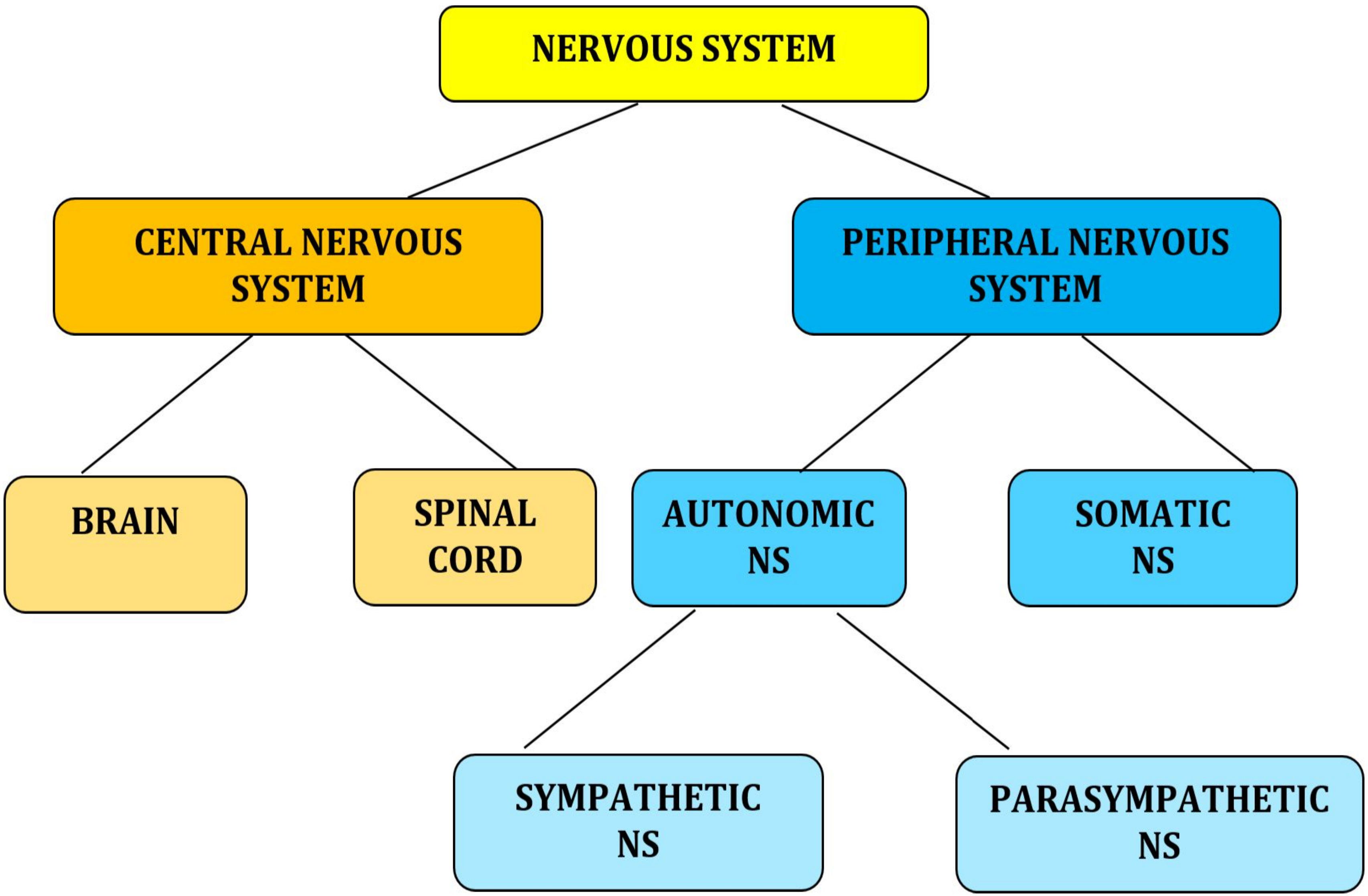
NERVOUS SYSTEM

- Highly organized system of human body.
- It is the organ system containing a network of specialized cells called neurons that coordinate the actions and transmit signals between different parts of the body.



ORGANIZATION OF THE NERVOUS SYSTEM



ORGANIZATION OF THE NERVOUS SYSTEM

1. Central Nervous System

- **the central nervous system containing the brain and spinal cord,**
- **It is the center of integration and control**

2. Peripheral Nervous System

- **the peripheral nervous system which is a network of nerves and neural tissues branching out throughout the body.**
- **Peripheral nervous system comprises of**
- **31 Spinal nerves**
- Carry info to and from the spinal cord
- **12 Cranial nerves** which carry

 Carry info to and from the brain

BASIC FUNCTIONS OF THE NERVOUS SYSTEM

1. **<u>Sensation:</u>**
   - Monitors changes/events occurring in and outside the body. Such changes are known as stimuli and the cells that monitor them are receptors.
2. **<u>Integration:</u>**
   - The parallel processing and interpretation of sensory information to determine the appropriate response
3. **<u>Reaction/Response:</u>**
   – Motor output.
   – response to information processed through stimulation of *effectors*
      - muscle contraction, glandular secretion

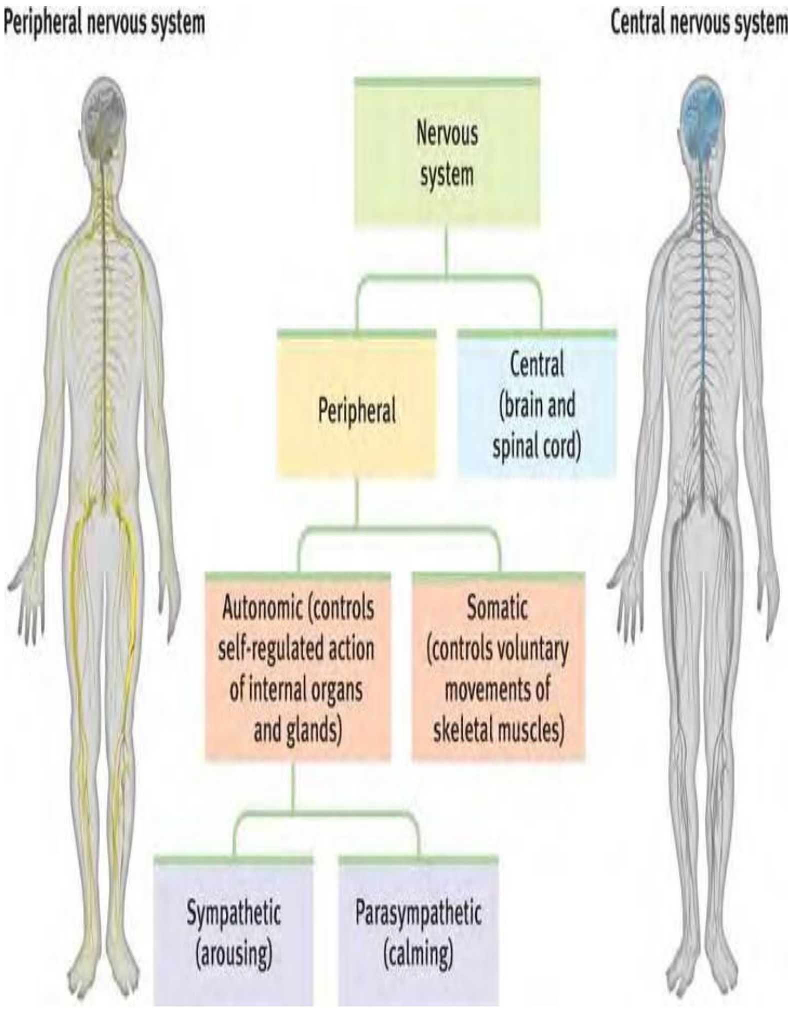
The activation of muscles or glands (typically via the release of neurotransmitters (NTs))

Together, the central nervous system (CNS) and the peripheral nervous systems (PNS) transmit and process sensory information and coordinate bodily functions.

The brain and spinal cord (the CNS) function as the control center.

They receive data and feedback from the sensory organs and from nerves throughout the body, process the information, and send commands back out.

Nerve pathways of the PNS carry the incoming and outgoing signals.



<u>CENTRAL NERVOUS SYSTEM</u>

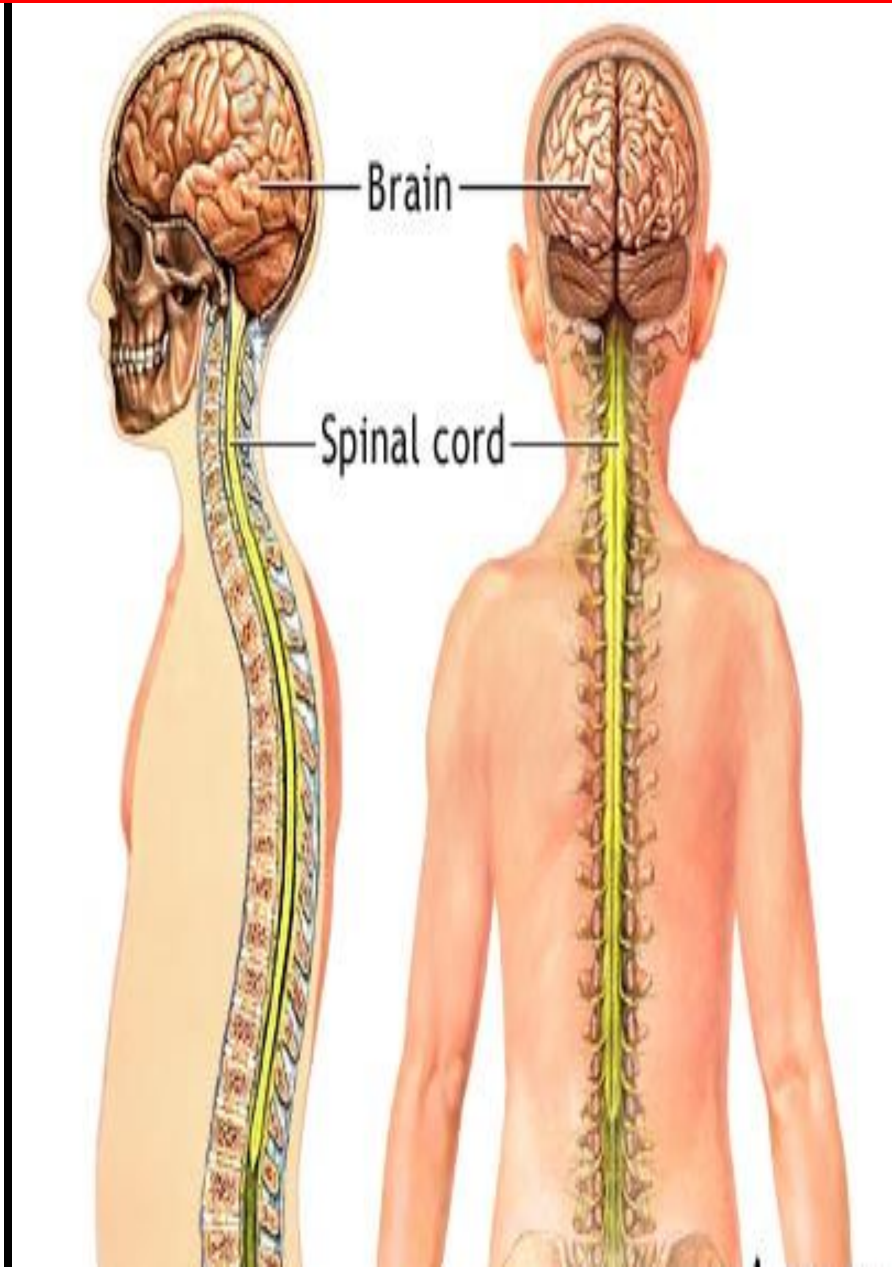
- Divided into two parts:
- ***<u>Central N.S:</u>***

  Consists of

  – **Brain,** contained within the cranium

  – **Spinal cord,** lodged in the vertebral canal

  – Two portions are continuous with one another at the level of the upper border of the atlas vertebra.



The spinal cord…

The spinal cord has 31 segments:

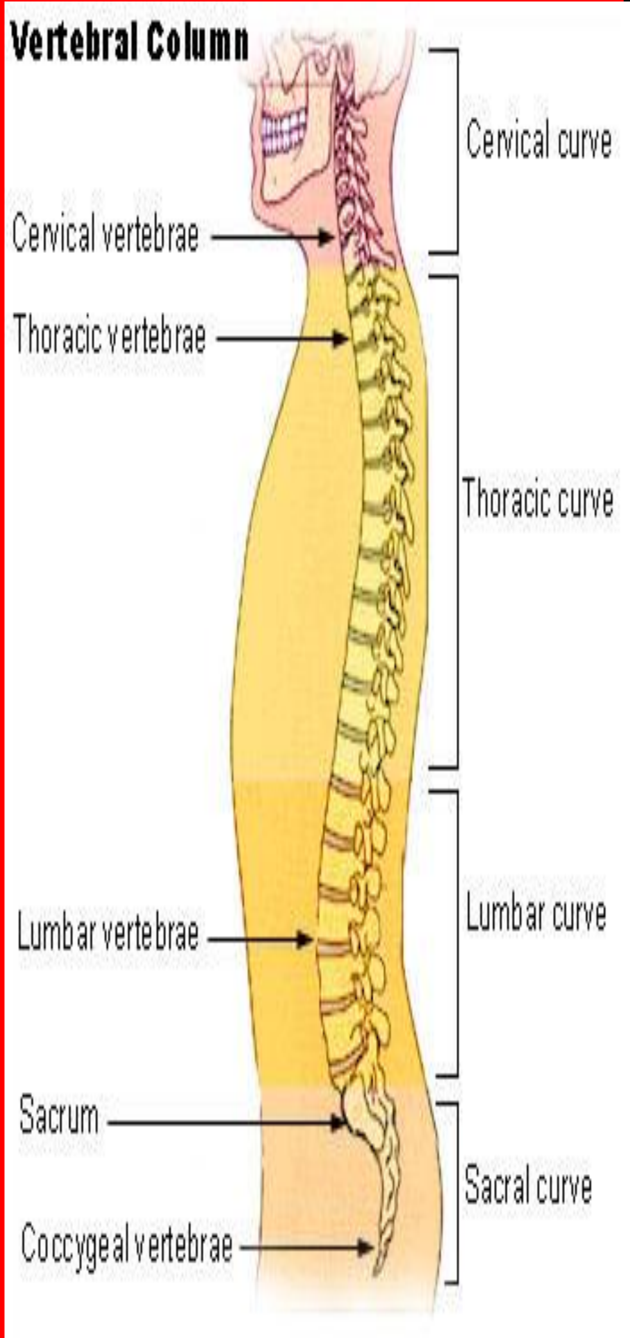
8 cervical segments that correspond to the C1-C8 vertebrae;

12 thoracic segments corresponding to the T1-T12 vertebrae;

5 lumbar segments corresponding to L1-L5 vertebrae,

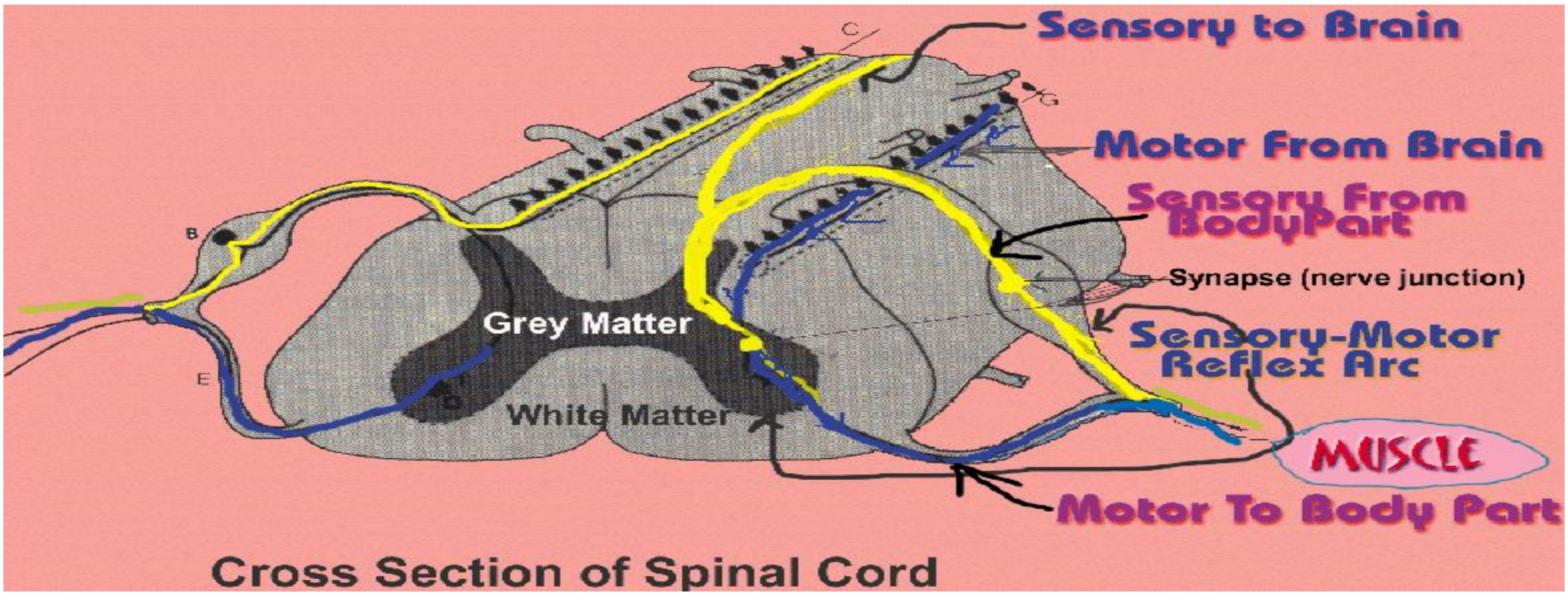
5 sacral segments corresponding to S1-S5 vertebrae, and 1 coccygeal (cock SĬJ ee ul) segment.

The spinal cord is about 44 cm long



The spinal cord…

At each segment of the spinal cord, left and right pairs of sensory and motor nerves branch out and connect to the peripheral nervous system. Impulses travel back and forth to the brain and back to the muscles.



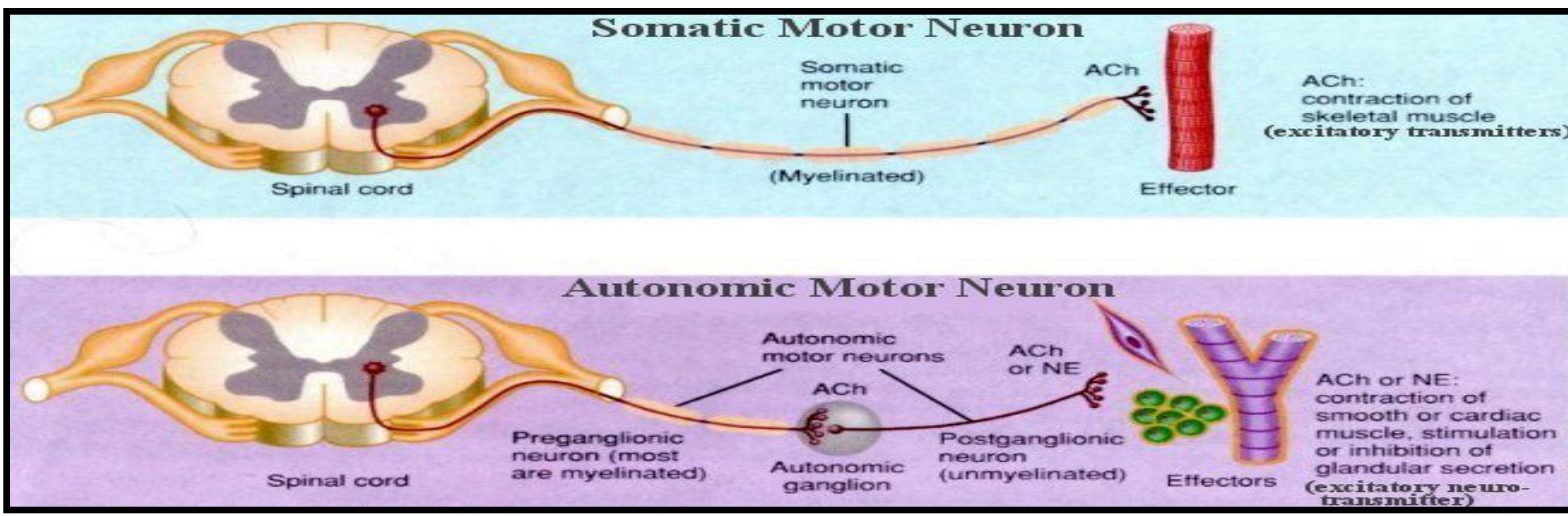
Peripheral nervous system

– ***Peripheral N.S:***

- It is a series of nerves by which the central nervous system is connected with the various tissues of the body.
- PNS is not protected by the bone of spine and skull, or by the blood-brain barrier, leaving it exposed to mechanical injuries.
- Divided into:

– Somatic nervous system

– Autonomic nervous system

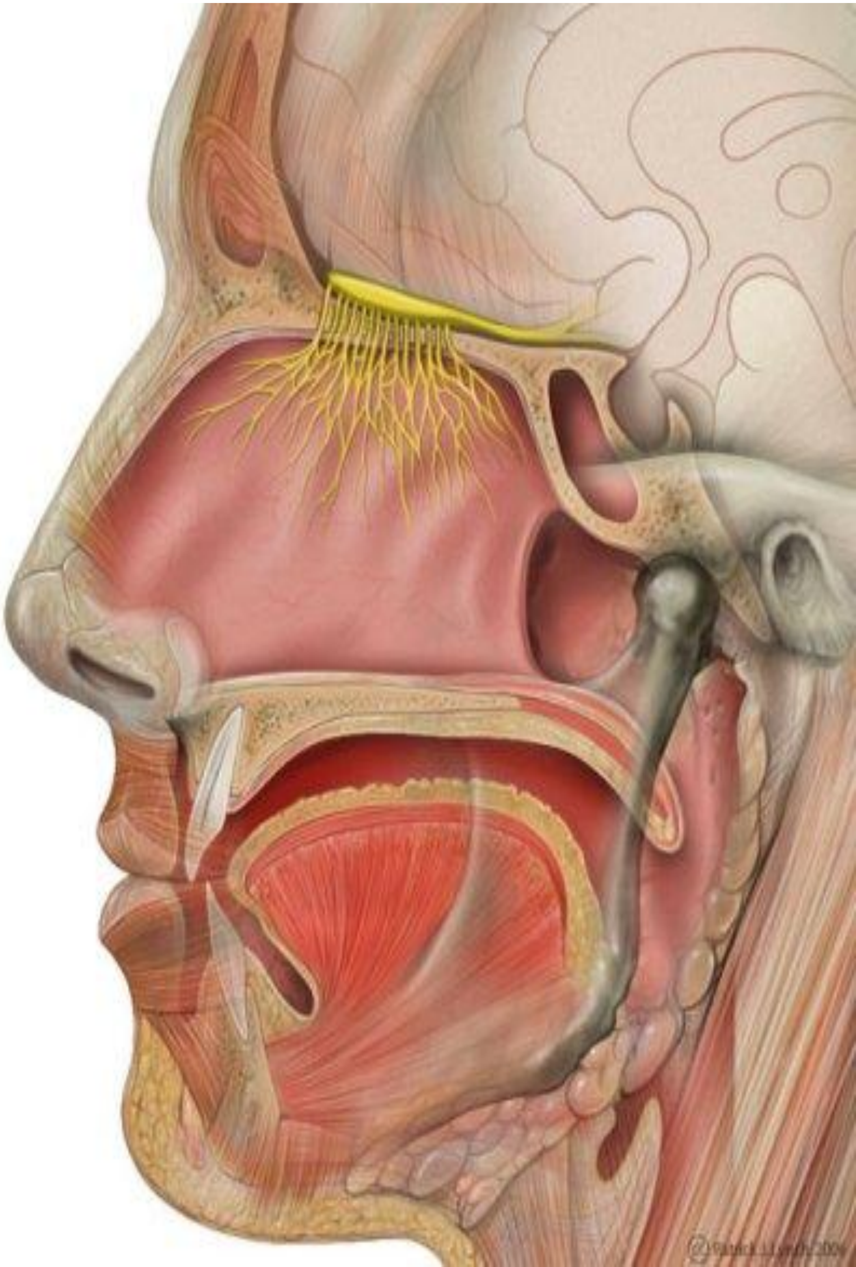


Peripheral nervous system…

The network of nerves branching out throughout the body from the brain and spinal cord is called the peripheral nervous system.

In addition to the 31 pairs of spinal nerves mentioned in the slides on the spinal cord, there are 12 pairs of cranial nerves that attach to the brain:

1- The olfactory nerve carries sensory input for smell



Peripheral nervous system…

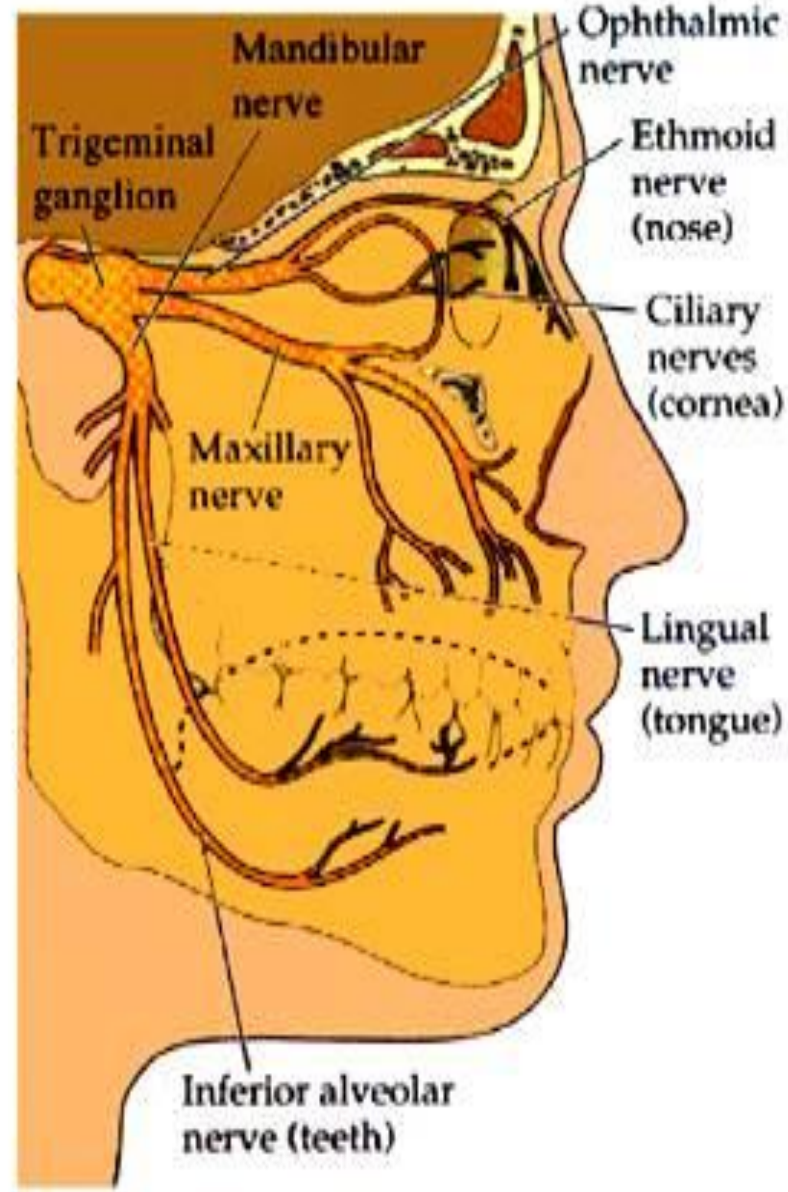
II. **The optic nerve carries sensory input for vision**

III. **The oculomotor nerve controls muscles of the eye and eyelid**

IV. **The trochlear nerve (TRŎK lee ur) controls the eyeball**

V. **The trigeminal nerve (try JEM ĭ nul) controls the face, nose, mouth, forehead, top of head, and jaw.**

VI. **The abducens nerve** (ab DŪ senz) also controls the eyeball



Peripheral nervous system…

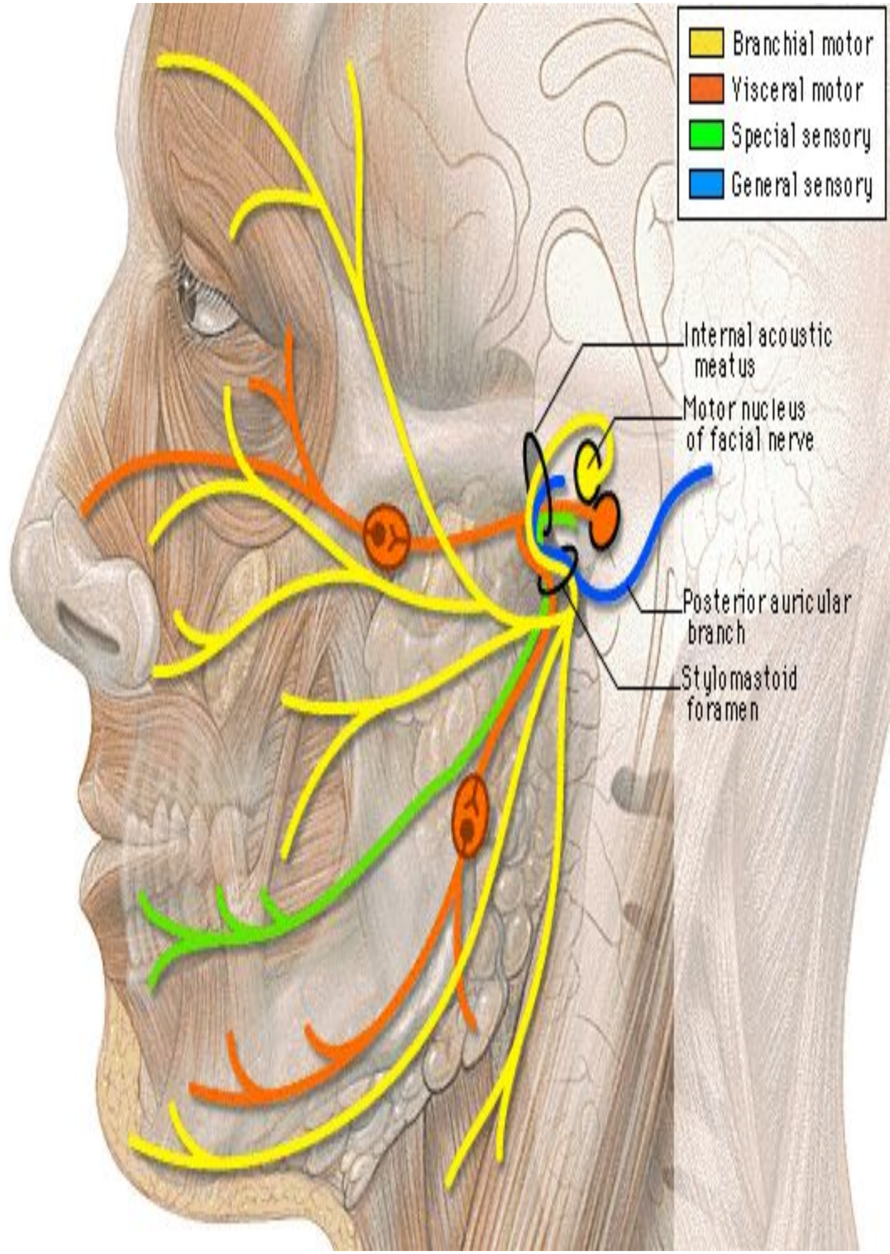
VII. **The facial nerve**

controls muscles of the face and
scalp, and part of the tongue for sense of taste.

VIII. **The auditory or cochlear nerve**

provides sensory input for hearing and equilibrium.

IX. **The glossopharyngeal** nerve controls saliva, swallowing, and taste.

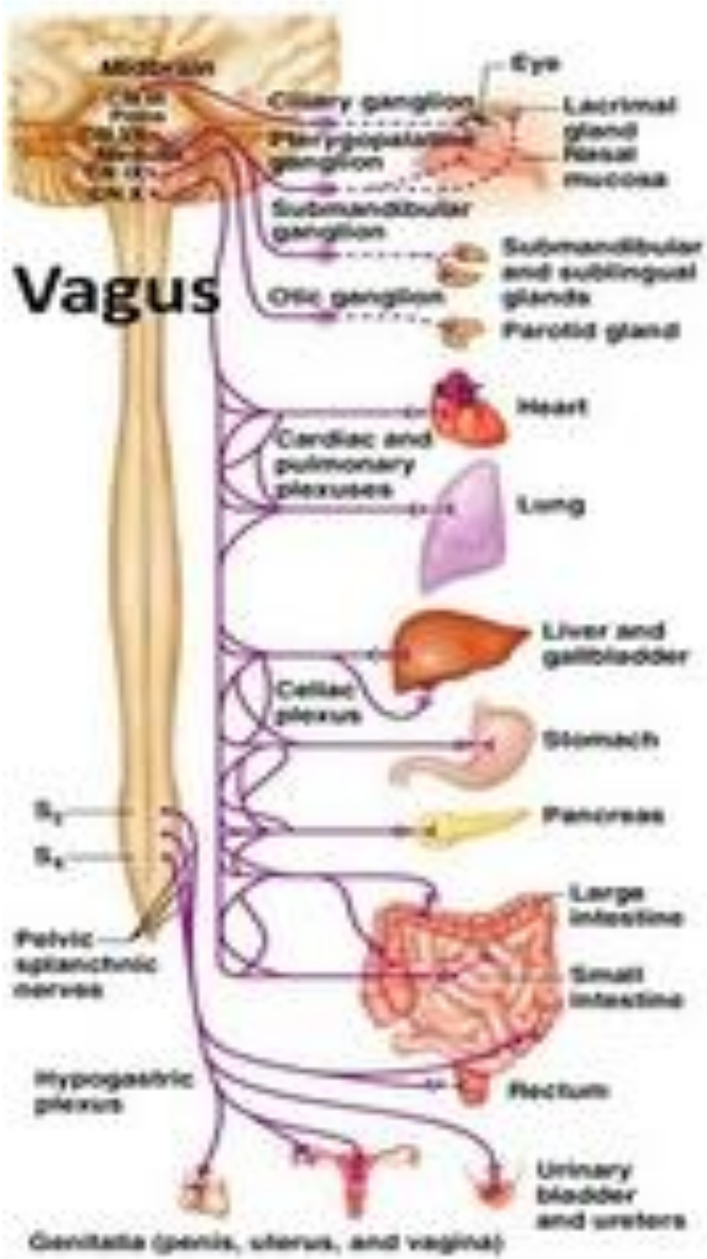


Peripheral nervous system…

X. The vagus (VĀ gus) nerve is the longest cranial nerve, extending to and controlling the heart, lungs, stomach, and intestines.

XI. The accessory nerve permits movement of the head and shoulders.

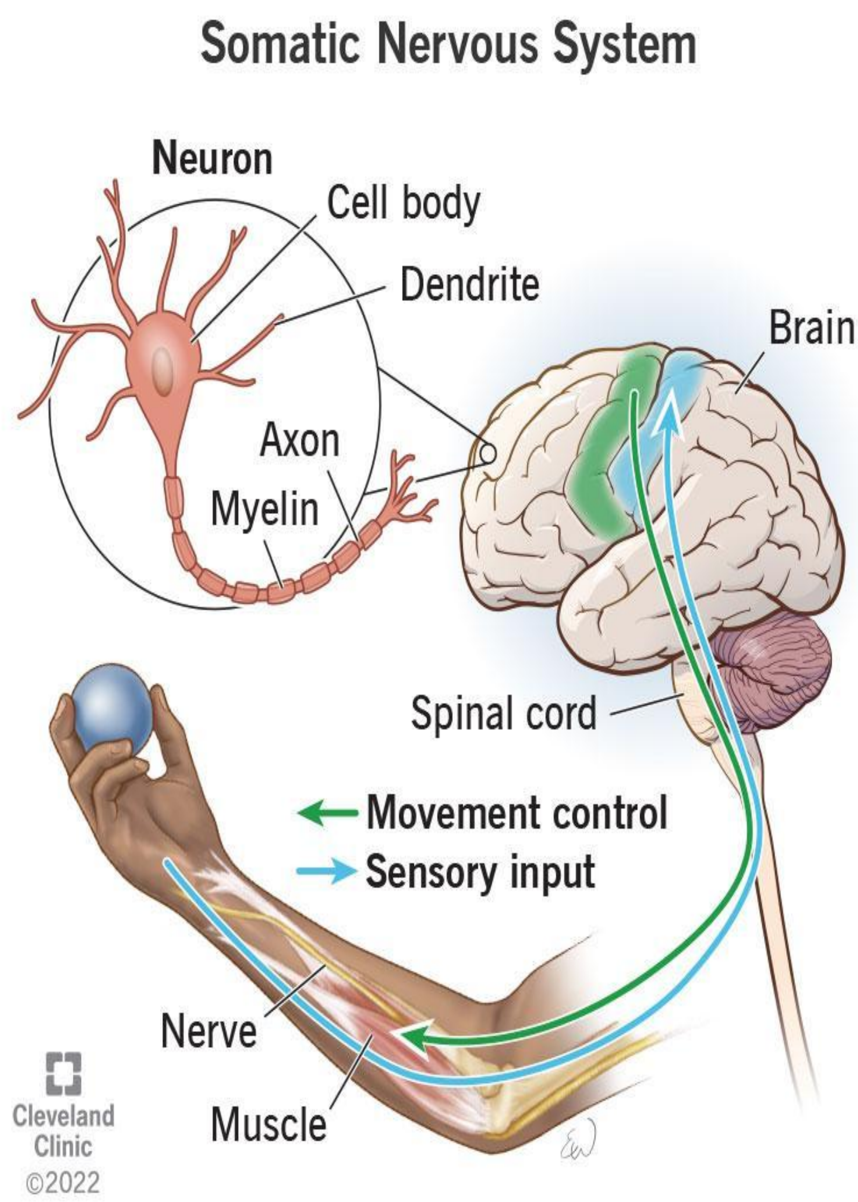
XII. The hypoglossal nerve (hī pah GLOSS ul) controls the muscles of the tongue.



SOMATIC NERVOUS SYSTEM

The somatic nervous system is the part of the peripheral nervous system responsible for processing sensory information and controlling voluntary movements.
It consists of nerves that link the central nervous system (brain and spinal cord) to the skin, sensory organs, and skeletal muscles, allowing you to consciously interact with the world through senses like touch, and actions like walking or writing

A substantial portion of the peripheral nervous system comprises 43 different segments of nerves (12 pairs of cranial and 31 pairs of spinal nerves), which help us perform daily functions.
The somatic nervous system consists of both afferent (sensory) and efferent (motor) nerves

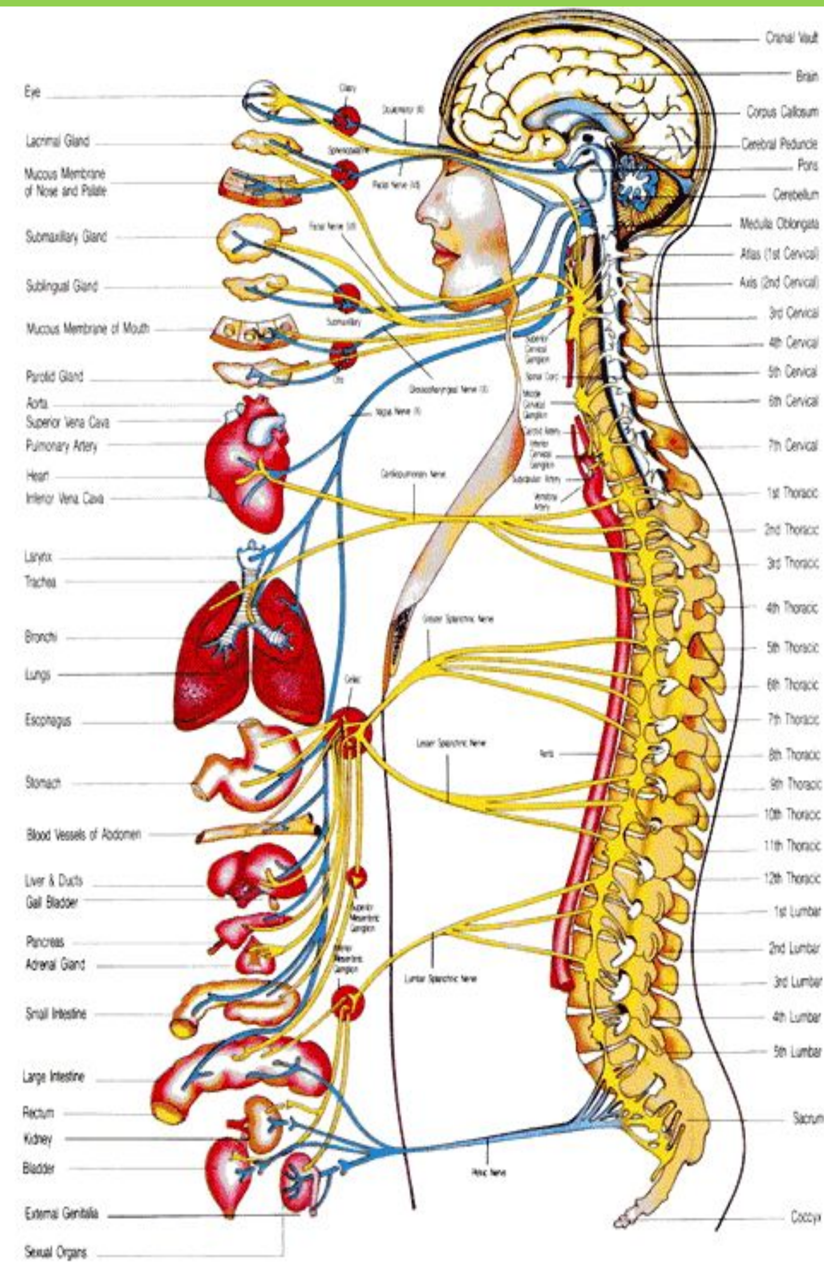


Autonomic nervous system…

The autonomic nervous system is a part of the peripheral nervous system..

It controls the involuntary bodily functions such as sweating, gland secretions, blood pressure, and the heart.

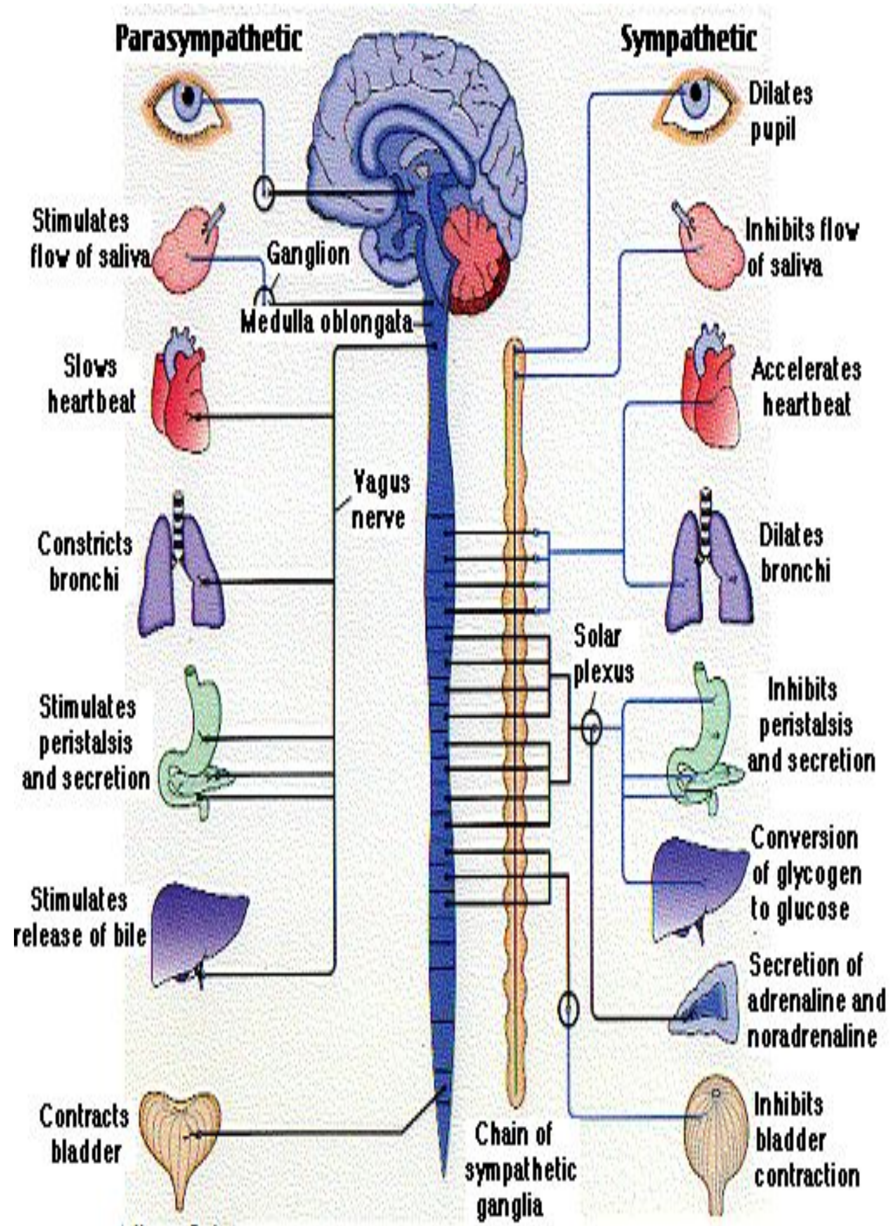
It is divided into the 'sympathetic' and 'parasympathetic' divisions.



Autonomic nervous system…

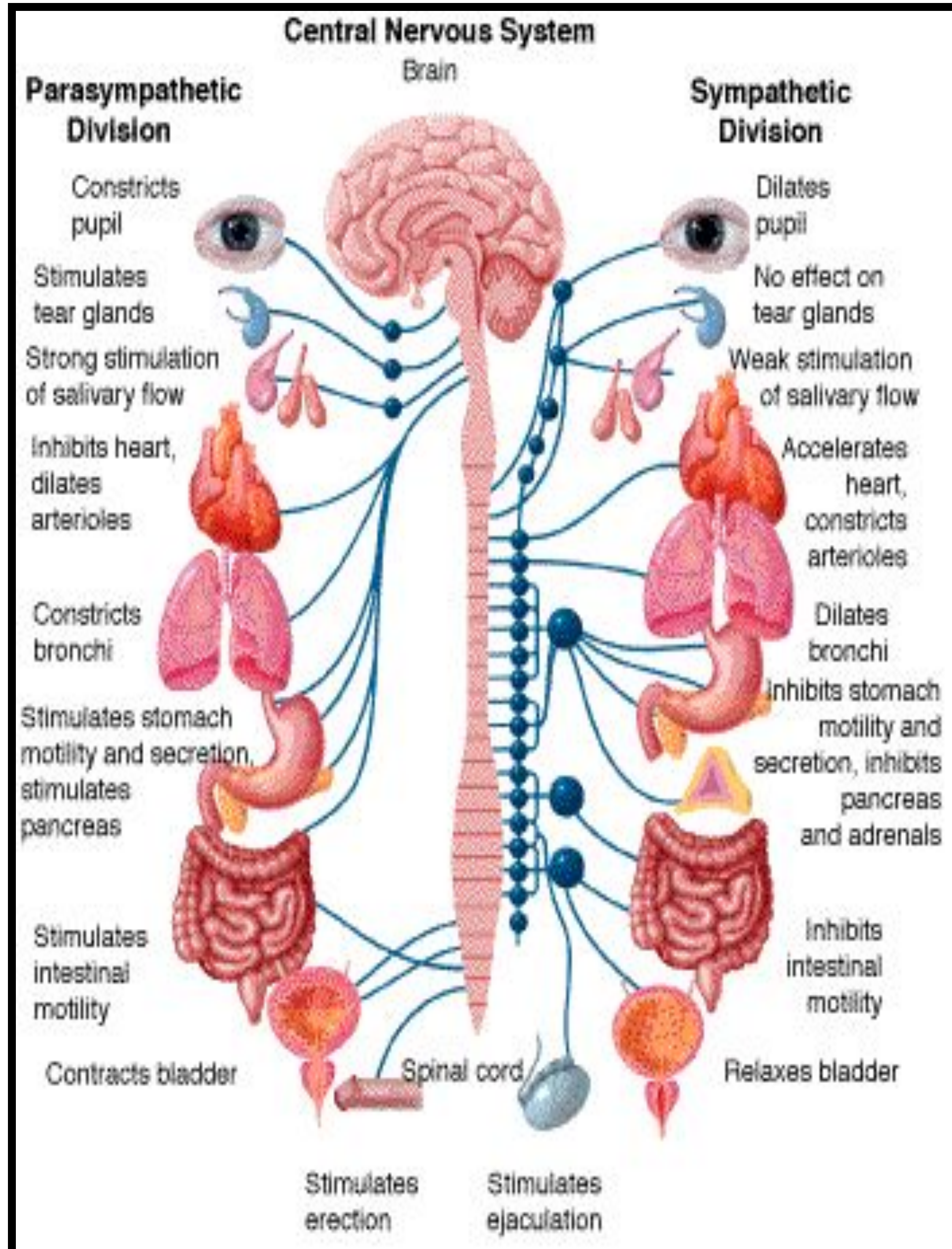
The sympathetic nervous system is responsible for the 'flight or fight' responses…i:e increased alertness, metabolic rate, respiration, blood pressure, heart rate, and sweating AND a decrease in digestive and urinary function.

The parasympathetic nervous system counteracts the responses of the sympathetic system… restoring homeostasis.



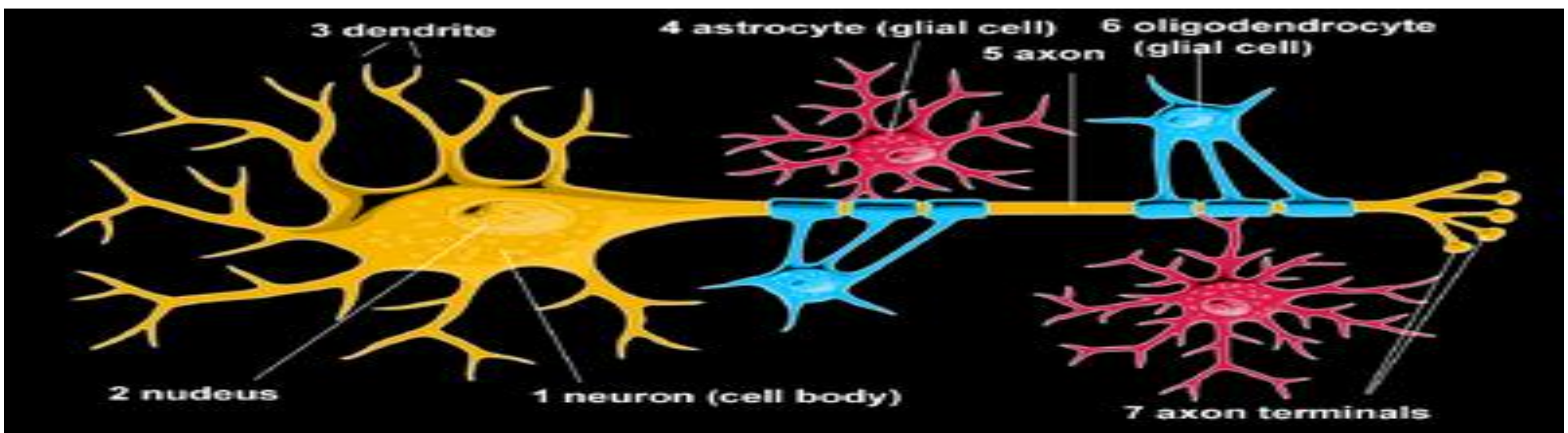
AUTONOMIC NERVOUS SYSTEM

- The ANS affects
- heart rate,
- digestion,
- respiration rate,
- salivation,perspiration, diameter of the pupils, micturition (urination),and sexual arousal.
- Where as most of its actions are involuntary, some, such as breathing, work in tandem with the conscious mind.



NERVOUS TISSUE

- Nervous Tissue comprises of
  Neurons,
  Neuroglia,
  and blood vessels
- **Neuron** is the basic structural and functional unit of nervous tissue
- **Neuroglia** Neuroglia, or glial cells, are non-neuronal supporting cells in the central and peripheral nervous systems that outnumber neurons and provide crucial metabolic, structural, and homeostatic support.

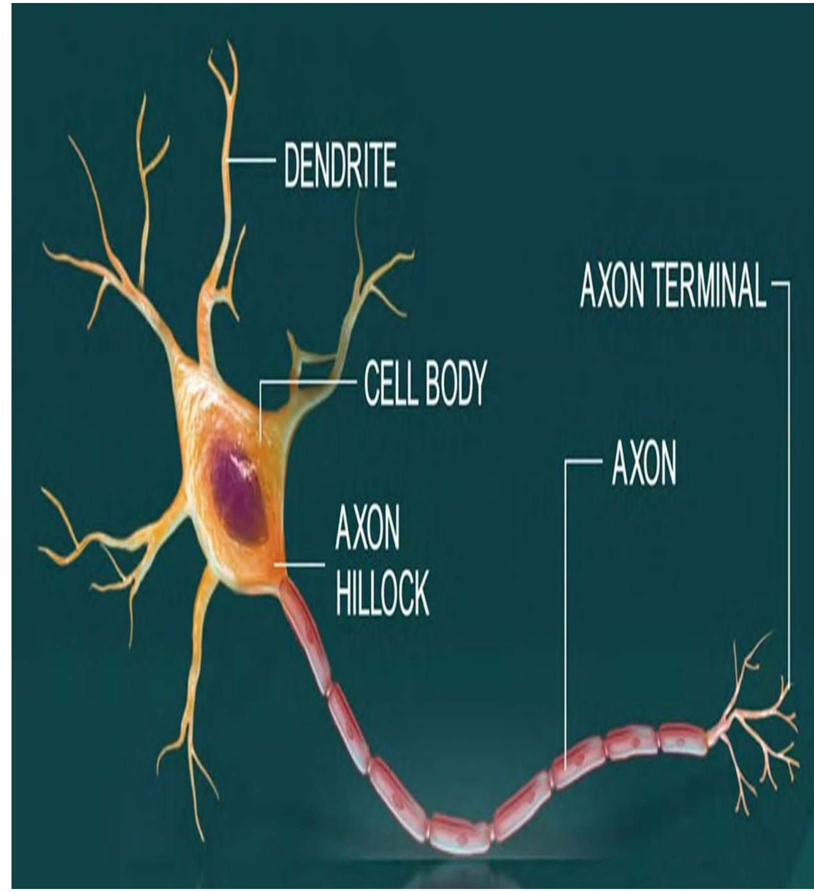


Neurons in Nervous Tissue Relay Rapid-Fire Signals.

All nervous tissue, from the brain to the spinal cord to the furthest nerve branch, includes cells called neurons.

Neurons are charged cells: they conduct electrical signals to pass information through the body.

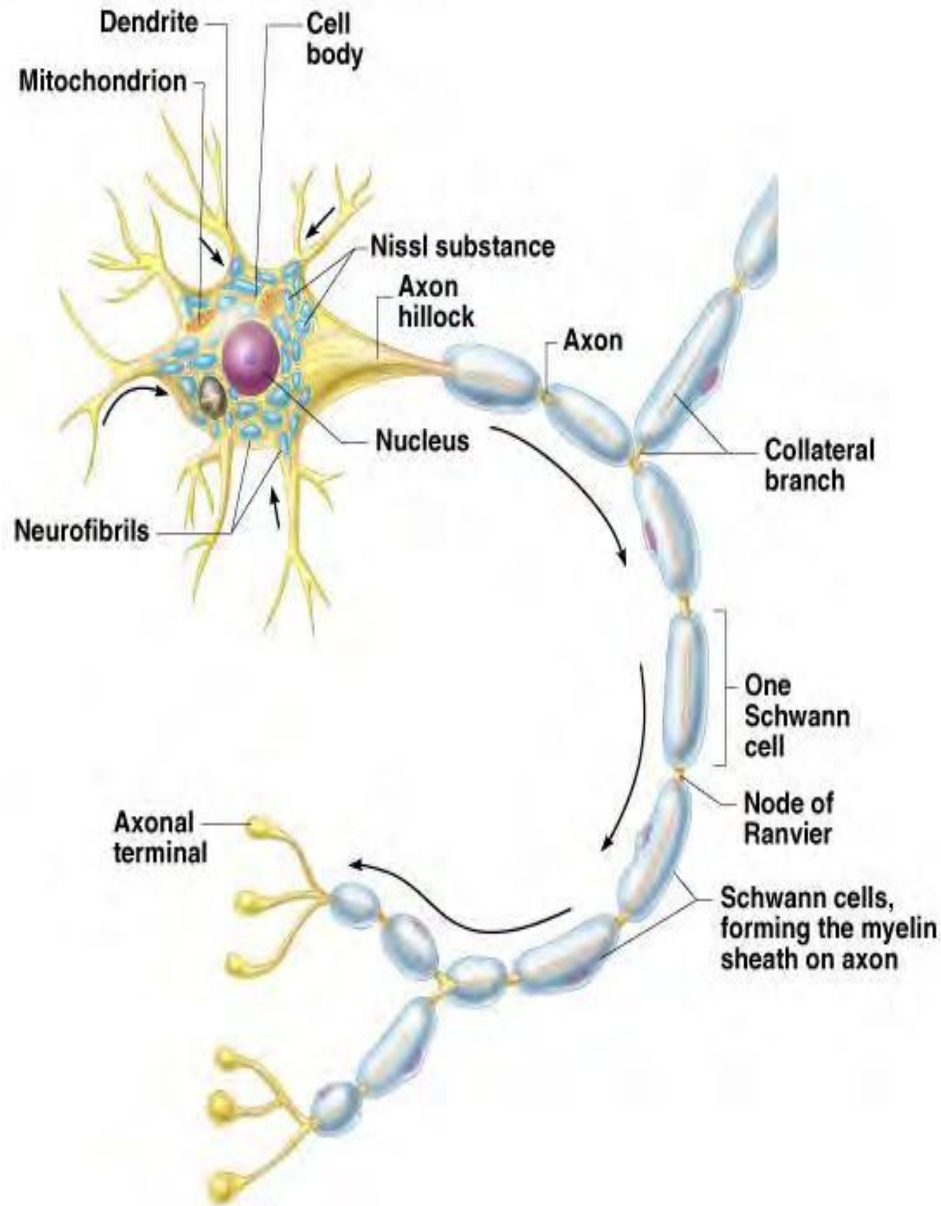
A typical neuron consists of a cell body, dendrites, and an axon with an axon terminal.



Neuron Anatomy

Extensions outside the cell body

- Dendrites conduct impulses toward the cell body
- Axons – conduct impulses away from the cell body



(a)

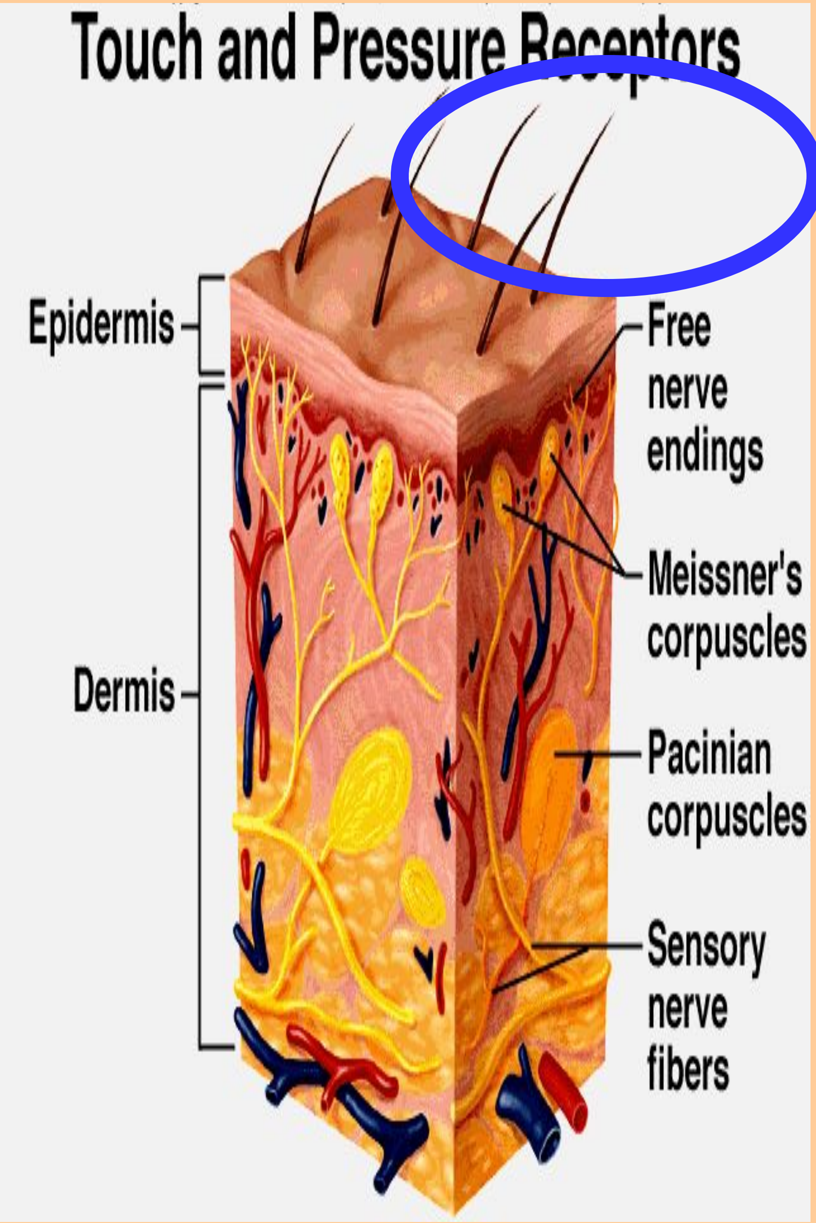
Types of neurons…

Neurons are the nerve cells, the structural and functional units of the nervous system.

They conduct impulses that enable the body to interact with its internal and external environments.

There are various types of neurons.

tissues that support the nerve cells is called neuroglia (nur ROH glee ah).



Neurons Based on Function

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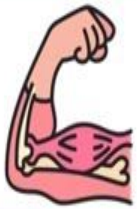
Sensory Neuron

These neurons detect stimuli from the environment, such as light, sound, and touch. They transmit this information to the central nervous system (CNS).

Visual neurons
touch neurons
Auditory neurons

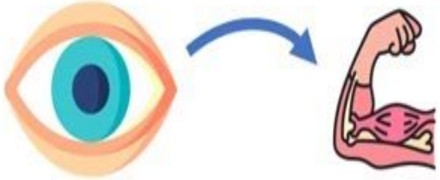
Motor Neuron



These neurons control the muscles and other organs of the body. They transmit signals from the CNS to the muscles, telling them to contract or relax.

Upper motor neurons
Lower motor neurons

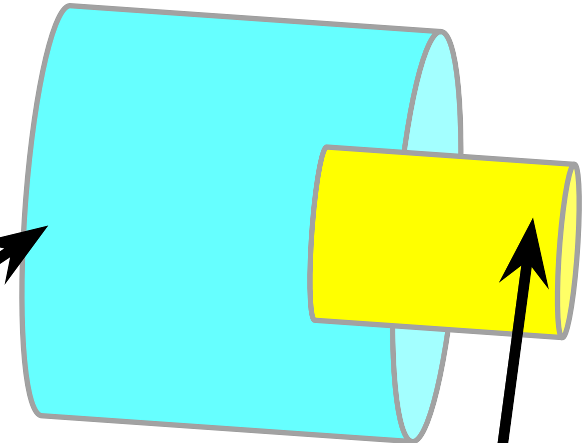
Interneuron



These neurons connect sensory neurons and motor neurons. They help to process information and coordinate the activity of different parts of the nervous system.

Nerve fibers…

Schwann cells, only found on peripheral nerves which can regenerate



Myelin sheath, a fatty layer of insulation on some nerve fibers

The axon that transmits the message

Because of it’s insulating properties, it keeps the electrical signal strong and also moving faster along the Axon, as compared to a Neuron without the Myelin, as shown in the diagram at the right.

There are no Schwann cells on nerve fibers in the central nervous system, therefore damage to those nerve fibers is not reversible. A bundle of nerve fibers is simply called ‘a nerve’.

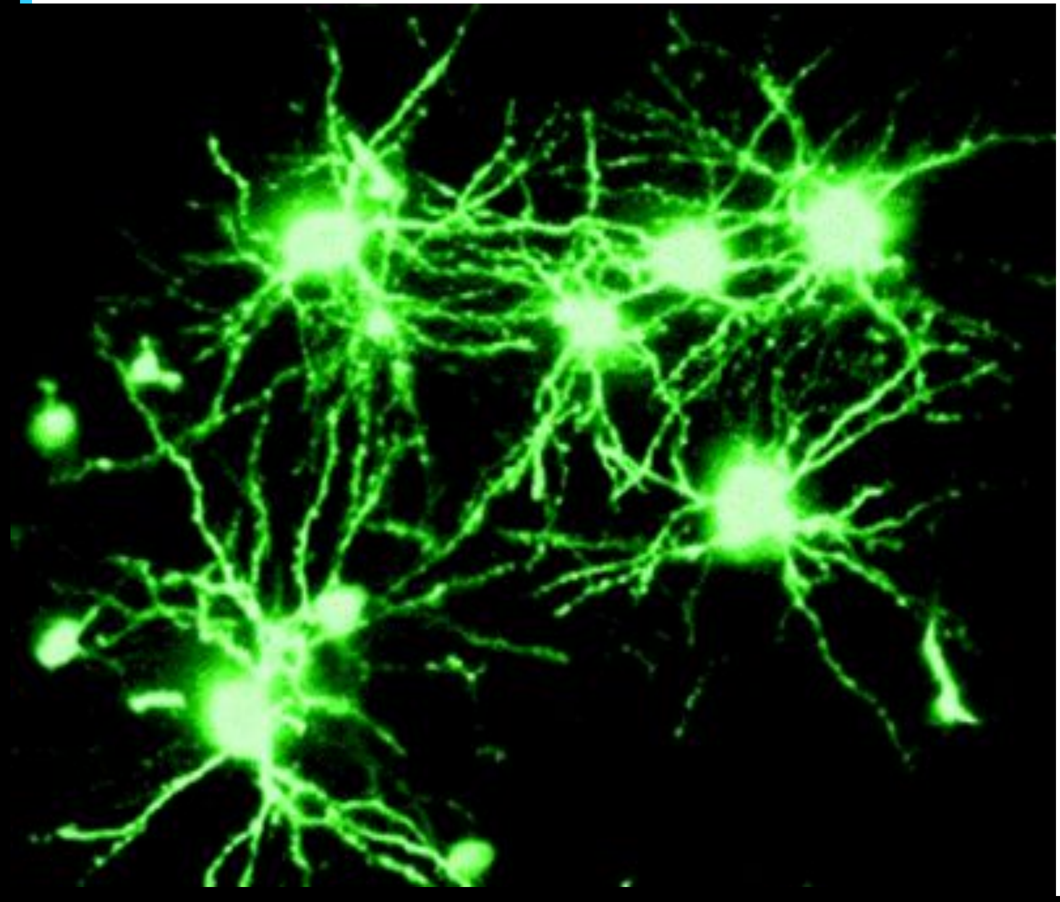
AFFERENT nerves conduct impulses to the central nervous system;

EFFERENT nerves conduct impulses to the muscles, organs, and glands.

Synapses…

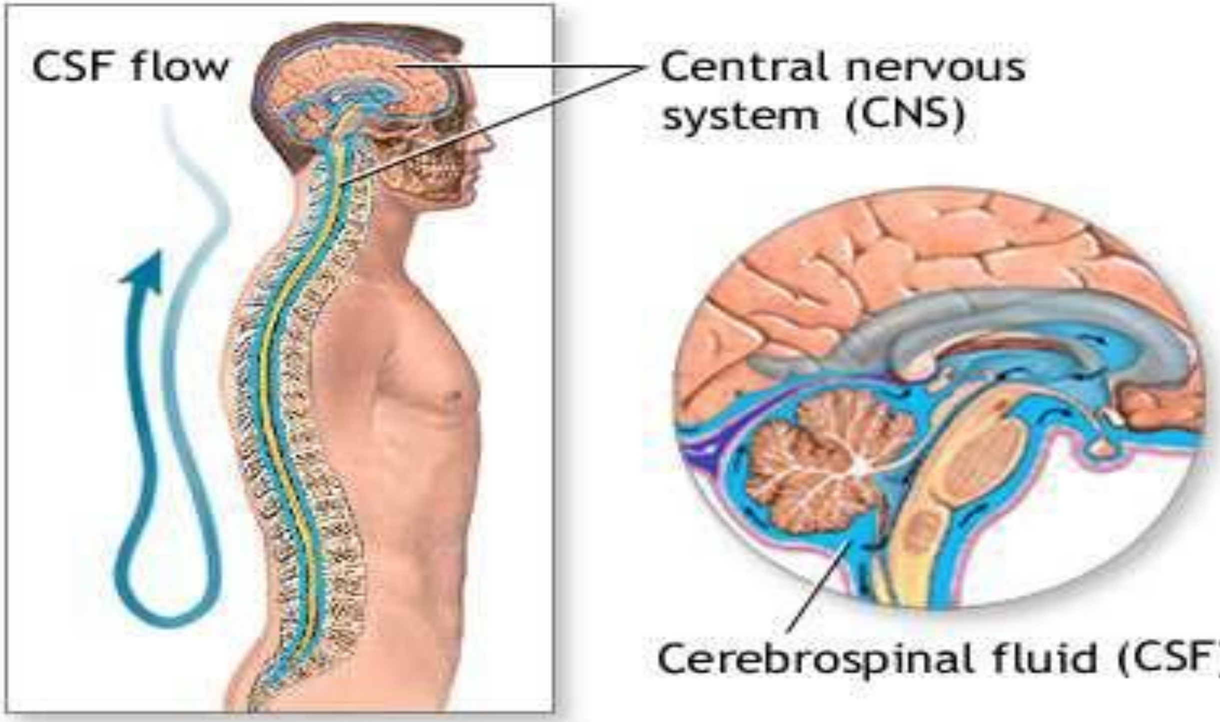
Nerve impulses are transmitted via branches called synapses.

The synapses are connectors… hooking dendrites and axons from one neuron to another.



The number of synapses influences transmission. That number can decrease with disease, lack of stimulation, drug use, etc.

Cerebrospinal fluid…



A colorless fluid is produced in the ventricles of the brain; it surrounds the brain and spinal cord. It is called cerebrospinal fluid, and it cushions the brain and cord from shocks that could cause injury. It is maintained at a level around 1/2 - 2/3 cup.